

Assignment 4 Solution

1. For R-type instruction, the funct3 field(14-12th bits) and funct7 (30th bit) define which R-type operation will take place, such as add, sub, xor, etc.
2. The zero pin outputs either 1 or 0 depending on the ALU result. It is used for conditional branching. For beq instruction, the registers are subtracted from each other, and if the ALU result is 0, the zero pin gives 1, meaning they are equal and hence jump to the branch target address.
3. Pipeline time = $40 \times 5 = 200\text{s}$
Single Cycle time = $20 + 20 + 20 + 40 = 100\text{s}$ (Memory operation not required for add instruction)
Therefore, single cycle will run it faster.

4. ***Ld X5, 10(X21)***

ALUSrc	RegWrite	MemRead	MemWrite	MemtoReg
1	1	1	0	1